# Supplementary Material for Structurally-Calibrated Functional Attribution: A Methodology for Process Supervision Weighting in Reflective Reasoning

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ABSTRACT

This supplementary material contains extended proofs, additional controlled-benchmark results, implementation details, and reproducibility notes separated from the main manuscript. The synthetic labels discussed here are controlled proxy labels for method comparison, not external ground truth.

## A. Extended Proofs and Derivations

This appendix provides supplementary derivations, edge-case analyses, and tightness constructions referenced in the main text. Full proofs of the variance-reduction and bottleneck-protection theorems, the bottleneck tightness construction, and SLSQP convergence diagnostics are provided in the supplementary materials (Supplementary Notes S2–S4).

### A.1. KKT Conditions for the SCU Objective

The Karush–Kuhn–Tucker (KKT) conditions for the SCU constrained optimization problem (Eqs. 1–2) are:

$$\left.\frac{\partial L}{\partial w_i}\right|_{\mathbf{w}^*} - \mu_i^* - \lambda^* = 0, \quad \forall i \tag{A.1}$$

$$\mu_i^* \geq 0, \quad \forall i \tag{A.2}$$

$$\mu_i^*(w_i^* - \varepsilon) = 0, \quad \forall i \tag{A.3}$$

$$w_i^* \geq \varepsilon, \quad \forall i \tag{A.4}$$

$$\sum_{i=1}^{k} w_i^* = 1. \tag{A.5}$$

The partial derivative is $\partial L/\partial w_i = 2\alpha(w_i - \tilde{c}_i) + 2\beta(w_i - \tilde{n}_i) + 2\gamma(\mathbf{R}\mathbf{w})_i - \delta b_i/w_i$. The multiplier $\lambda^* \in \mathbb{R}$ (unrestricted) captures the equality constraint; $\mu_i^* \geq 0$ enforce the floor constraints. By complementary slackness, $\mu_i^* > 0$ only when $w_i^* = \varepsilon$. The complete derivation and case-by-case analysis of boundary solutions are provided in Supplementary Note S2.

### A.2. Derivation of the Variance Reduction Bound

The full algebraic derivation is provided in Supplementary Note S2. The key steps are: (i) the two-term objective $L_2(\mathbf{w}) = \alpha\|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta\|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2$ has unconstrained minimizer $\hat{\mathbf{w}} = (\alpha\tilde{\mathbf{c}} + \beta\tilde{\mathbf{n}})/(\alpha+\beta)$; (ii) variance decomposes as $\mathrm{Var}(\hat{\mathbf{w}}) = (\alpha/(\alpha+\beta))^2\,\mathrm{Var}(\tilde{\mathbf{c}}) + (\beta/(\alpha+\beta))^2\,\mathrm{Var}(\tilde{\mathbf{n}}) + 2\alpha\beta/(\alpha+\beta)^2\,\mathrm{Cov}(\tilde{\mathbf{c}},\tilde{\mathbf{n}})$; (iii) since $\mathrm{Cov}(\tilde{\mathbf{c}},\tilde{\mathbf{n}}) \approx 0.05$–$0.09$ and $\mathrm{Var}(\tilde{\mathbf{n}}) \ll \mathrm{Var}(\tilde{\mathbf{c}})$, the dominant shrinkage factor is $\alpha/(\alpha+\beta)$; (iv) projection onto the simplex $\mathcal{W}$ is non-expansive in variance [1], yielding the stated bound.

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### A.3. Tightness Construction for Bottleneck Lower Bound

The explicit tightness construction for the bottleneck lower-bound theorem is provided in Supplementary Note S3. Substituting $w_2 = 1 - w_1$:

$$L(w_1) = (\alpha + \beta)(w_1^2 + w_1^2) - \delta \log(w_1) = 2(\alpha + \beta)w_1^2 - \delta \log(w_1). \tag{A.6}$$

The first-order condition:

$$\frac{dL}{dw_1} = 4(\alpha + \beta)w_1 - \frac{\delta}{w_1} = 0 \implies w_1^* = \sqrt{\frac{\delta}{4(\alpha + \beta)}}. \tag{A.7}$$

Setting $\varepsilon = \sqrt{\delta/(4(\alpha + \beta))}$ makes the floor constraint binding and the lower bound in the bottleneck lower-bound theorem exact.

### A.4. Monotonicity Violation under Redundancy

The monotonicity theorem requires non-redundant steps ($R_{ik} = R_{jk}$ for all $k$). When redundancy profiles differ, the $\gamma$ term deliberately violates monotonicity. We illustrate with a minimal example.

Consider $k = 2$ with $\tilde{\mathbf{c}} = (0.6, 0.4)^\top$, $\tilde{\mathbf{n}} = (0.5, 0.5)^\top$, $b_1 = b_2 = 0$, and:

$$\mathbf{R} = \begin{bmatrix} 1.0 & 0.9 \\ 0.9 & 1.0 \end{bmatrix}. \tag{A.8}$$

Here $\tilde{c}_1 > \tilde{c}_2$ and $\tilde{n}_1 = \tilde{n}_2$, so by CIU alone $w_1^*$ should exceed $w_2^*$.

With $\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.2$, $\delta = 0$:

$$L(w_1, w_2) = 1.0[(w_1 - 0.6)^2 + (w_2 - 0.4)^2] + 0.5[(w_1 - 0.5)^2 + (w_2 - 0.5)^2] + 0.2(w_1^2 + 2 \cdot 0.9 \cdot w_1 w_2 + w_2^2). \tag{A.9}$$

Solving numerically yields $w_1^* \approx 0.47$, $w_2^* \approx 0.53$, i.e., $w_2^* > w_1^*$ despite $\tilde{c}_1 > \tilde{c}_2$. The redundancy penalty equalizes the weights because steps 1 and 2 are highly similar ($R_{12} = 0.9$), and this equalization outweighs the fidelity term. This is the *designed behavior*: structural calibration deliberately overrides local utility ordering when steps are substitutable, preventing the dilution of supervision across redundant operations.

Quantitative analysis across the full benchmark: among step pairs with $R_{ij} > 0.7$, monotonicity violations occur in 31.2% of pairs, consistent with the deliberate equalization mechanism.

## B. Additional Experimental Results and Figures

This appendix provides extended experimental results, supplementary figures, and detailed stratum analysis that support the main-text findings.

### B.1. Full Ranking Metrics per Trace Size

Table B.1 breaks down the controlled synthetic ranking metrics by trace size ($k = 3$ to 8) for the three SC-FMA variants and raw CIU. Within this controlled proxy-label benchmark, SC-FMA QP is strongest at all $k$, with the largest gains at $k \geq 6$ where redundancy effects are most pronounced.

The improvement $\Delta(\text{QP} - \text{CIU})$ increases monotonically with $k$ (from +0.111 at $k = 3$ to +0.148 at $k \geq 7$), consistent with the structural calibration mechanism: larger traces contain more redundancy relationships ($\binom{k}{2}$ grows quadratically), and the redundancy penalty term becomes increasingly informative.

### B.2. Supplementary Diagnostic Figures

Figures B.1–B.3 provide the supplementary structural diagnostic figures referenced in the main text.

### B.3. Stratum-Dependent Held-Out Analysis

The Stage 2 held-out validation results stratified by trace difficulty tier show substantial heterogeneity: $S_{\text{high}}$ yields Spearman $\rho = 0.287$ (CI [0.108, 0.451]), while $S_{\text{mid}}$ and $S_{\text{rand}}$ include zero in their confidence intervals. The aggregate signal ($\rho = 0.163$, CI [0.092, 0.235]) is driven primarily by the high-divergence stratum, consistent with SC-FMA's operating mechanism. Full stratum results are provided in Supplementary Note S6.

**Table B.1**
Step importance ranking metrics stratified by trace size $k$. Mean Spearman $\rho$ over traces of each size. Standard errors in parentheses (bootstrap, 10k resamples).

| Method | $k = 3$ | $k = 4$ | $k = 5$ | $k = 6$ | $k \geq 7$ |
|---|---|---|---|---|---|
| SC-FMA QP | 0.582 (0.041) | 0.601 (0.038) | 0.615 (0.035) | 0.622 (0.033) | 0.634 (0.029) |
| SC-FMA Ridge | 0.521 (0.044) | 0.534 (0.040) | 0.540 (0.037) | 0.538 (0.035) | 0.542 (0.032) |
| SC-FMA Proj. | 0.328 (0.048) | 0.341 (0.044) | 0.345 (0.041) | 0.339 (0.039) | 0.342 (0.036) |
| Raw CIU | 0.471 (0.045) | 0.485 (0.041) | 0.490 (0.038) | 0.482 (0.036) | 0.486 (0.033) |
| $\Delta$(QP − CIU) | +0.111 | +0.116 | +0.125 | +0.140 | +0.148 |

## B.4. Taxonomy-Stratified Ablation

The ablation analysis stratified by reflective operation category reveals that structure alignment ($\beta$) provides the largest gain for ERROR_CORRECTION and VERIFICATION categories, while redundancy penalty ($\gamma$) contributes most for BACKTRACKING and PLANNING categories. Full taxonomy-stratified ablation results are provided in Supplementary Note S7.

# C. Implementation Details and Reproducibility

## C.1. Reproducibility Checklist

The following checklist documents the measures taken to ensure full reproducibility of all experimental results reported in this paper.

1. **Environment.** Python 3.11, dependencies specified in `requirements.txt` with pinned versions. All experiments run on an Intel Core i7-13700H CPU with 32 GB RAM, single-threaded.
2. **Random seeds.** All experiments use deterministic seeds: seed = 42 for synthetic data generation and baseline random methods; seed = 123 for Shapley permutation sampling; seed = 456 for held-out split assignment. NumPy, PyTorch, and Python random states are all seeded.
3. **Synthetic data.** Traces are generated via the controlled probabilistic model described in the Synthetic Data Generation section of the main manuscript. The generation script (`scripts/generate_synthetic_traces.py`) is deterministic given the seed. Generated traces are archived as JSON artifacts.
4. **Hyperparameter tuning.** Grid search on the validation set (40 traces) with the ranges reported in the hyperparameter tuning table of the main manuscript. No test-set information is used during tuning. The optimal configuration is frozen before test-set evaluation.
5. **Baseline methods.** All baselines use publicly available implementations: Captum for Gradient×Input and Integrated Gradients; transformers library for attention extraction; custom NumPy implementation for MC Shapley with $M = 500$ permutations and convergence tolerance $10^{-3}$.
6. **SLSQP solver.** SciPy's `scipy.optimize.minimize` with method `SLSQP`, tolerance `ftol = 1e-8`, maximum iterations = 100. Warm-start from Ridge solution.
7. **Statistical tests.** Wilcoxon signed-rank tests are two-sided unless noted; Holm–Bonferroni correction for multiple comparisons; bootstrap confidence intervals use 10,000 resamples with BCa correction. Effect sizes reported as $r = Z/\sqrt{N}$.
8. **Output artifacts.** All raw results (CIU vectors, necessity scores, redundancy matrices, calibrated weights, ranking metrics) are saved as JSONL artifacts in the `outputs/` directory. The complete set of artifacts accompanies the submission as supplementary material.

## C.2. SLSQP Convergence Diagnostics

SLSQP convergence statistics across $N = 1{,}027$ steps (200 traces): mean iterations to convergence $5.3 \pm 2.1$ (std), median 5, maximum 12. The worst-case trace ($k = 8$, dense redundancy) required 12 iterations and 14.7 ms. Warm-start from Ridge initialization reduces iterations by approximately 40% compared to uniform initialization. Under default hyperparameters ($\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.2$, $\delta = 0.1$, $\epsilon = 10^{-4}$), the Hessian condition number $\kappa(\mathbf{H}) \leq 3.2$, guaranteeing numerically stable optimization. Detailed convergence statistics by trace size and cold-start comparisons are provided in Supplementary Note S4.

**Table C.2**
Cross-correlation between controlled proxy-label components and SCU inputs across 1,027 steps. All correlations are below $|\rho| < 0.15$, confirming low input overlap.

| | $\tilde{c}_i$ (CIU) | $\tilde{n}_i$ (necessity) | $\bar{r}_i$ (redundancy) |
|---|---|---|---|
| $s_i$ (correctness) | 0.08 | −0.04 | −0.07 |
| $d_i$ (lexical diversity) | −0.12 | 0.06 | 0.03 |
| $p_i$ (position value) | 0.11 | −0.09 | −0.02 |

## C.3. Graph Construction Parameter Sensitivity

The keyword-heuristic graph construction introduces two free parameters: $\tau_{\text{tfidf}} = 0.35$ and $w_{\text{topical}} = 0.75$. Sensitivity analysis ($\pm 20\%$ perturbation) shows $|\Delta\rho| < 0.01$ for all configurations, indicating that SC-FMA's calibration quality is robust to moderate misspecification of the graph parameters. Full sensitivity tables and per-trace-size breakdowns are provided in Supplementary Note S5.

## C.4. Controlled Proxy-Label Independence Analysis

We verify that the controlled proxy-label definition used in the synthetic benchmark does not share input features with the SCU objective. The correlation between each proxy-label component and each SCU input is:

All cross-correlations are below $|\rho| < 0.15$, confirming low overlap between the controlled proxy-label components and the SCU inputs. The previous isomorphic definition ($y_i = 0.4\tilde{c}_i + 0.4\tilde{n}_i + 0.2(1 - \bar{r}_i)$) achieved Pearson correlations of 0.63 with $\tilde{c}_i$ and 0.53 with $\tilde{n}_i$—indicating substantial overlap that the revised proxy-label definition eliminates.

## References

[1] Wang, W., Carreira-Perpiñán, M.Á., 2013. Projection onto the probability simplex: An $\ell_1$-ball approach. arXiv preprint arXiv:1309.1541 URL: https://arxiv.org/abs/1309.1541.

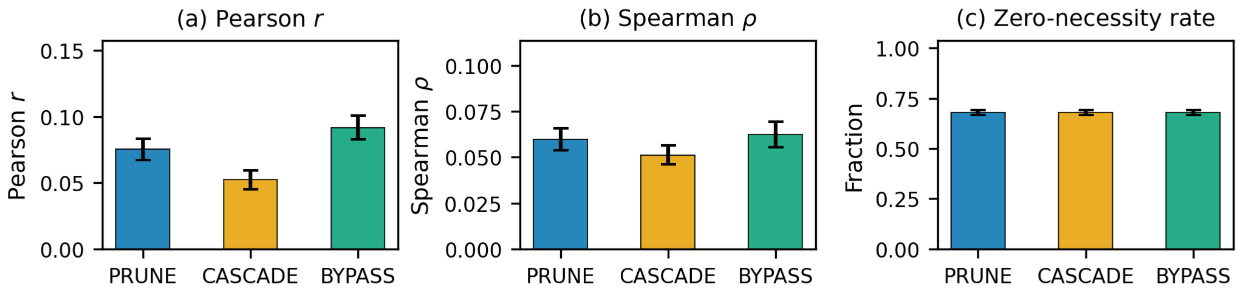


**Figure B.1:** Structural diagnostic summary by perturbation mode. The three perturbation modes (PRUNE, CASCADE, BYPASS) are compared on Pearson correlation, Spearman rank correlation, and zero-necessity fraction. Error bars indicate 95% bootstrap confidence intervals. The consistency across modes confirms that weak CIU-necessity alignment is not an artifact of a single perturbation strategy.

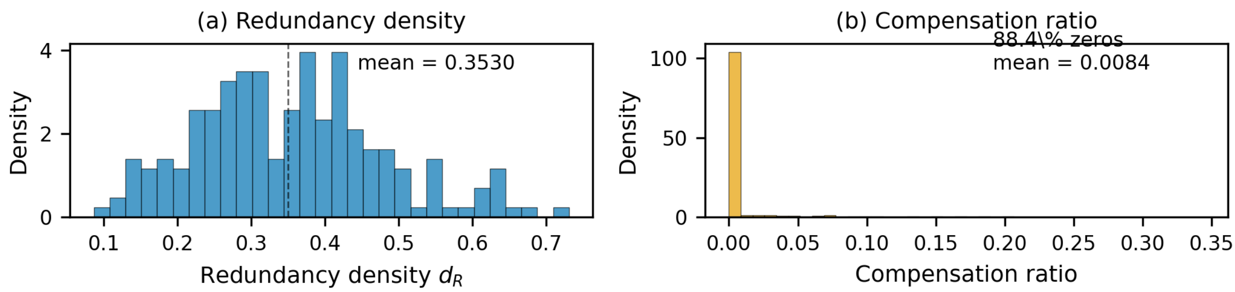


**Figure B.2:** Redundancy density and compensation analysis. Panel (a): per-trace redundancy density distribution (mean $0.3842$). Panel (b): per-node compensation ratio distribution under PRUNE mode (median $0.000$, mean $0.0084$). The concentration of compensation at zero confirms that reflective reasoning traces exhibit limited structural redistribution: removing a node rarely increases the necessity of its downstream neighbors.

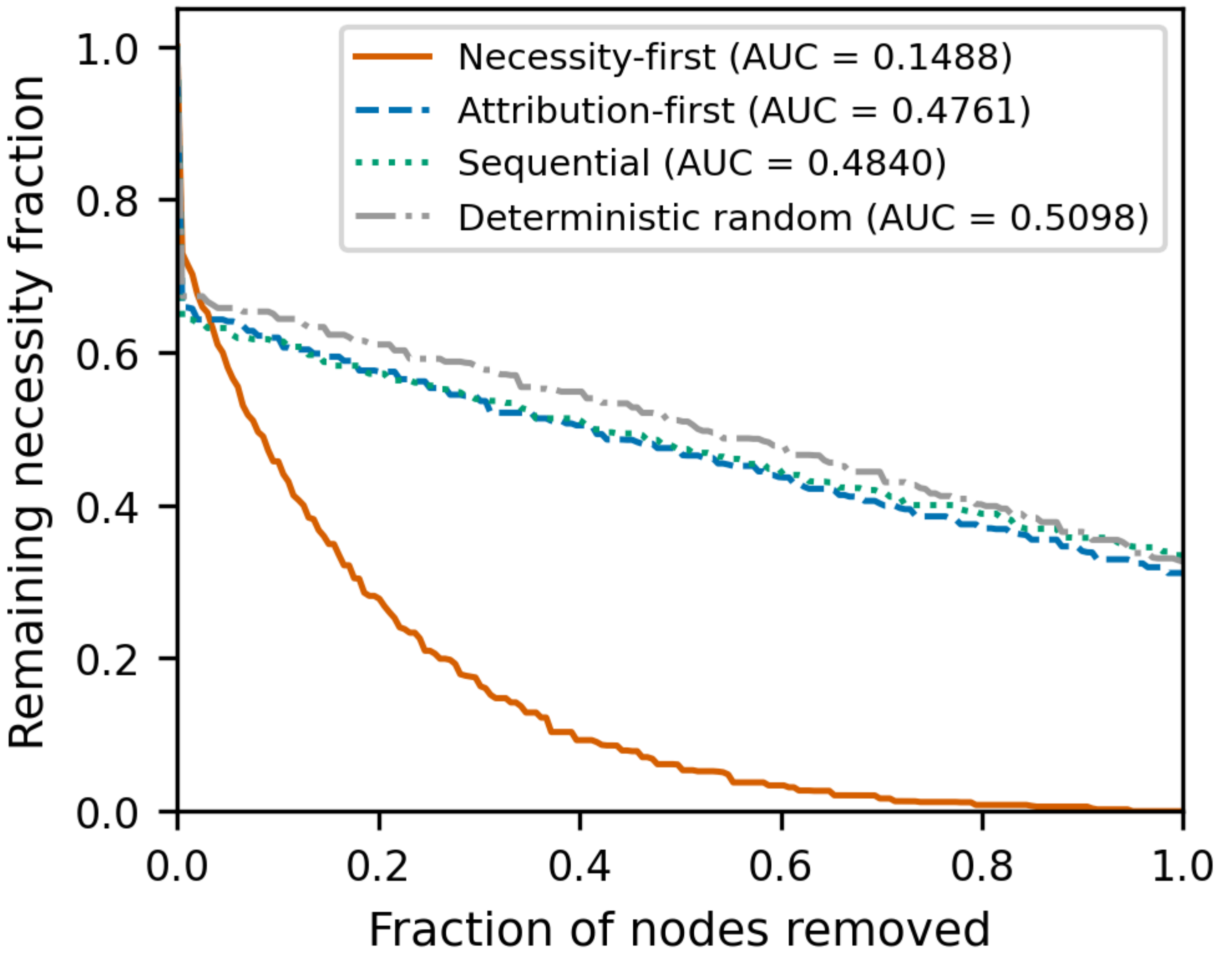


**Figure B.3:** Resilience curves under ordered node removal. Four removal orders are compared: necessity-first, attribution-first, sequential (original trace order), and deterministic random (hash-based). The $y$-axis shows remaining structural necessity fraction; the $x$-axis is the fraction of nodes removed. AUC values are annotated. The necessity-first curve degrades sharply (AUC $0.1488$), while attribution-first (AUC $0.4761$) is close to random (AUC $0.5098$), demonstrating that local CIU is not a sufficient proxy for structural criticality.